

**Exploring the Impact of Race, Gender, and Mental Health on the Magnitude of Victims in
American Mass Shootings: A Study on Gun Violence**

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Abstract

This investigation looked into how different factors influence the number of victims in mass shootings. These factors included the perpetrator's race, gender, and mental health status. This issue was investigated using five different machine learning regression models: Linear Regression, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Least Absolute Shrinkage and Selection Operator (LASSO), and Random Forest. With an MAE score of roughly 5, SVM proved to be the most accurate model. This model was then used to determine which "type" of person would cause the most amount of harm if they were to carry out a mass shooting. The most lethal mass shooting would be committed by a white woman with known mental health difficulties, according to an algorithm that tested possible combinations of characteristics. Conversely, a woman of unknown race and unknown mental health difficulties is expected to cause the fewest number of injuries. A permutation importance test was also performed to determine which variable has the largest impact on the number of victims. It was discovered that having a mental disorder increases the number of victims more than anything else.

Keywords: mass shootings, machine learning, permutation importance

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Introduction

It is not a mystery that the United States has one of the highest gun death and mass shootings rates in the world. Each year, tens of thousands of lives are lost in the United States due to firearms. Over the past decades, the number of mass shootings in the United States has skyrocketed. A "mass shooting" is defined by the FBI as any incident in which at least four persons are killed with a gun (U.S. Department of Justice, 2013). Where once there was only one shooting every couple of months, there are now hundreds every year; in 2020, 2021, and 2022, there were nearly double the number of mass shootings than there were days in the year (Yousif, 2023). There were 610 mass shootings in 2020, 690 in 2021, and 647 in 2022. Given these numbers, it is vital to understand the factors that cause a mass shooting to be deadly. This research is going to delve into how the race, gender, and mental health status of a perpetrator influences how many people that the person would harm if they were to commit a mass shooting. Various machine learning regression algorithms such as Linear Regression, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Least Absolute Shrinkage and Selection Operator (LASSO), and Random Forest will be used in order to assess the role these factors play. To test the results, metrics such as mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE) will be implemented to evaluate the accuracy of the models.

Literature Review

The topic of mass shootings is extremely popular today, and the amount of information about it is increasing as the number of shootings are rising. There are countless papers, articles, and datasets on the internet that discuss mass shootings and the factors that influence their

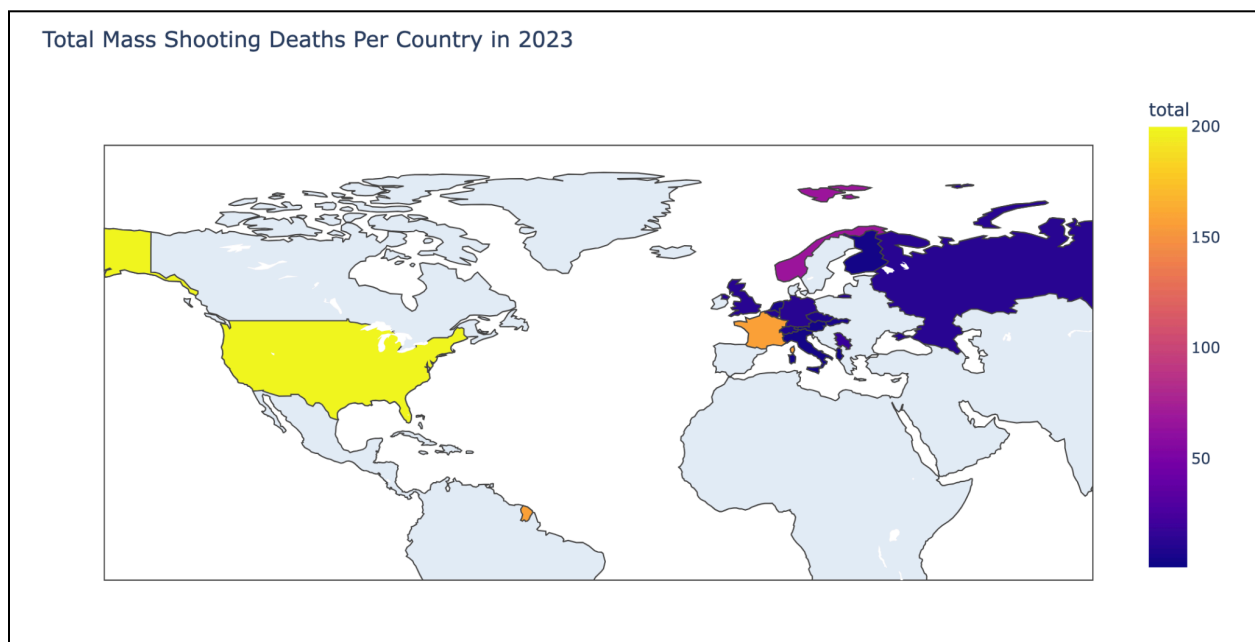
lethality and magnitude. This paper is going to focus only on the rapid increase of mass shootings per year in the United States, and how the race, gender, and mental health status of a person affects the number of victims (injured and dead) there would be if they were to commit a mass shooting.

Mass shootings in other countries compared to the United States

Research suggests mass shootings in the United States have consistently been higher than other countries for many years. A study done in 2016 found that the United States death rate by mass shootings was 1.5 for every one million people, compared to Switzerland's 1.7 per million people and 3.4 per million people in Finland. However, the study also found that those statistics were based on 133 mass shootings in the United States, two in Finland, and only one in Switzerland (Fisher and Keller, 2017). This demonstrates the magnitude of mass shootings in the United States compared to other countries.

Figure 1

Total Mass Shooting Deaths Per Country in 2023



Note. The United States has had the highest number of mass so far in 2023 compared to any other country.

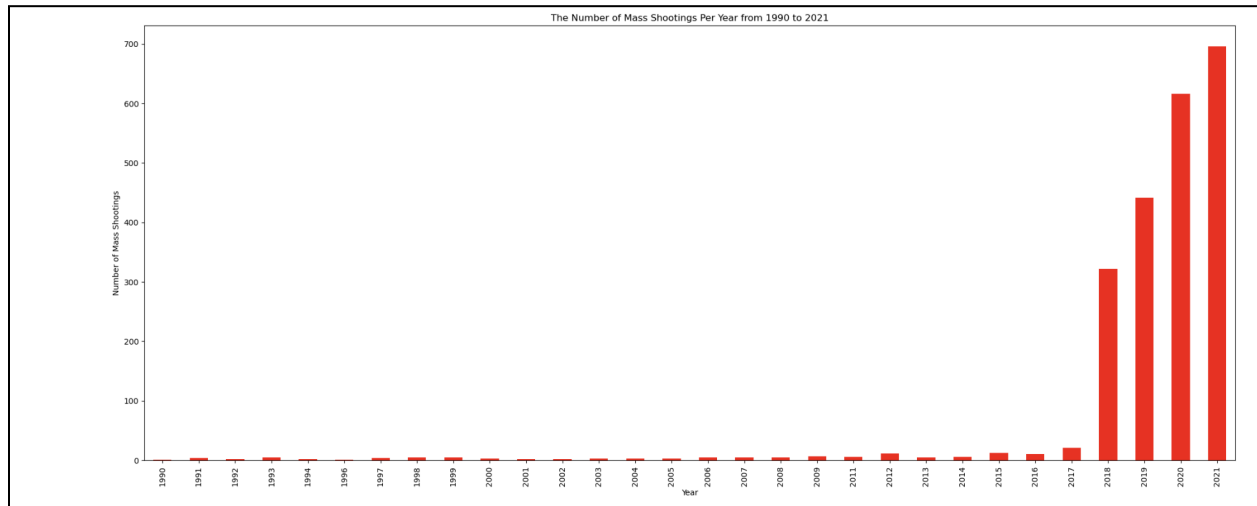
As figure 1 suggests, the United States has had the highest number of mass shootings so far in 2023 compared to any other country. It has had 199 mass shootings, and France is in second with 158 (World Population Review, 2023). Prior research makes it abundantly clear that the United States has substantially more mass shootings than any other country.

The reasons behind the rapid increase of mass shootings in the United States

Over the past decade, mass shootings in the United States have skyrocketed. The overall number of gun-related deaths in the United States in 2019 was 33,599. The number of deaths increased by 31% to 44,290 in 2022. 54% percentage of these deaths are firearm suicides, followed by homicides and mass shootings (Yousif, 2023). There has also been an increase in active shooter incidents over time as well. Active shooter incidents are when a singular person or multiple people are attempting to kill or are actively killing people in a heavily populated area. Between 2000 and 2021, the FBI discovered a huge increase in active shooter occurrences. In the year 2000, there were three incidents, but by 2021, the number had risen to 61 (Gramlich, 2023).

Figure 2

The Number of Mass Shootings Per Year from 1990 to 2021



Note. The data suggest that there has been a huge spike in mass shootings over time. The x-axis is labeled as “Year”, and the y-axis is labeled as “Number of Mass Shootings”.

Based on the information in figure 2, it is evident that there has been a huge spike in mass shootings over time. The dataset that was used to generate the model contains accounts of mass shootings ranging from 1900 to 2021, with a noticeable spike in 2018. Overall, there is strong data to support the fact that mass shootings have been increasing in the United States, and they are trending towards a higher increase in the future (Yousif, 2023).

Materials & Methods

When gathering the materials and datasets, the Kaggle website was used, which gave in-depth datasets regarding mass shootings and variables surrounding them. These variables included the race, gender, mental health status of the assailant, location, and many others. In the programming language Python, the irrelevant variables were removed from the dataset so that it was fit for running the desired model. Again, this model was designed to assess how a perpetrator's race, gender, and mental health condition may affect the number of victims they

might cause in a mass shooting. Once the dataset contained only the necessary variables, multiple machine learning regression models were run in order to predict how those factors affect the victim count in a mass shooting. The models ultimately used were Linear Regression, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Least Absolute Shrinkage and Selection Operator (LASSO), and Random Forest. In order to assess the accuracy of the models, metrics such as mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE) were used. The Linear Regression model had an MAE of ~ 9 , KNN with a value of around 8.4, SVM with a value of around 5.3, LASSO with a value of around 8.8, and Random Forest with a value of around 8.5. From these results, it was found that the SVM test was the most accurate model.

Material Gathering

As stated, all datasets were obtained from the Kaggle website, which is the largest data science community in the world, and lets users discover and publish datasets online (Kaggle). The dataset that was used for the machine learning models was a dataset of nearly 400 recorded mass shootings from 1966 to 2017. All datasets on Kaggle including the ones used in this research are openly accessible and do not require a license to be used.

Exploratory Data Analysis and Dataset Cleaning

Before the data was cleaned for the just necessary variables, there were multiple columns of data that were not going to be used in the machine learning models. The dataset included the titles of the shootings, location, date, summary, fatalities, injured, latitude, and longitude which were not ultimately used. Using Python, the unnecessary variables were dropped from the dataset, leaving only total victims, and the race, gender, and mental health issues of the perpetrator. In the race, gender, and mental health issue columns, there were still many

inconsistencies among how the values were listed. For example, a white person could either have been listed as “White”, “White American or European American”, or “White American or European American/Some other Race” under the race column. These inconsistencies in the values would have made the machine learning models less accurate and more convoluted overall. After all of the values were transformed so they were consistent and cohesive, the categorical data needed to be converted into numerical data for the models to run. Once that was completed, the dataset was cleaned and ready to be trained and tested in the machine learning models.

Figure 3

Cleaned Dataset Containing Only Total Victims, Mental Health Issues, Race, and Gender

	A	B	C	D
1	Total victims	Mental Health Issues	Race	Gender
2		573 Unknown	Unknown	Unknown
3		5 Yes	Asian	Male
4		3 Unknown	White	Male
5		5 Unknown	Unknown	Male
6		3 Yes	White	Male
7		3 Unknown	Black	Male
8		11 Yes	Latino	Male
9		5 Yes	Unknown	Male
10		6 Yes	Black	Male
11		16 Unknown	Black	Male
12		102 Unknown	Other	Male
13		4 Unknown	Unknown	Unknown
14		5 Unknown	Unknown	Unknown
15		6 Unknown	Black	Male
16		4 Unknown	Black	Male
17		5 Unknown	Unknown	Unknown
18		4 Unknown	White	Male
19		4 Yes	White	Male
20		4 Unknown	Unknown	Unknown
21		4 Unknown	White	Male
22		4 Unknown	Unknown	Unknown
23		5 Unknown	Unknown	Unknown
24		3 Unknown	Asian	Male
25		3 Unknown	Black	Male
26		3 Unknown	Black	Male
27		3 Unknown	Unknown	Male
28		3 Yes	White	Male
29		3 Unknown	Unknown	Male
30		3 Unknown	White	Male
31		3 Unknown	White	Male
32		3 Yes	White	Male
33		5 Unknown	Unknown	Unknown
34		4 Unknown	Black	Male
35		5 Unknown	Unknown	Unknown
36		4 Unknown	Unknown	Unknown
37		4 Unknown	Unknown	Unknown

Note. This cleaned version of the dataset was able to be used to be trained and tested by the machine learning models.

Machine Learning Modeling

In order to predict how gender, race, and mental health status of an individual affects how many people they would harm in a mass shooting, a regression machine learning model was needed. When variables in a dataset are numerical, regression models are used. However, when the variables fall into different non-numerical categories, categorical models are used. Five different machine learning regression models were used in this case. Before running them, the dataset was split into the training set and the test set, which ensures that the model is trained with the necessary information to make predictions, and tests that training on another part of the original dataset (GiniMachine). Once the training and test data was defined, the different machine learning models were trained and tested using Python. The first model tested was Linear Regression, as it is the most common regression model. This trial proved that the cleaned dataset was able to train and test a model, so four more models were tested as well: K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Least Absolute Shrinkage and Selection Operator (LASSO), and Random Forest.

Results

Out of all five regression models, SVM was the most accurate model, with an MAE of 5.289. This means that the SVM model was the best at predicting how a shooter's race, gender, and mental health might affect the number of victims if they were to carry out a mass shooting. Most of the other models are hovering around a 8-9 MAE, making the SVM the most accurate and best predictive model of all. To evaluate the models, MAE was the main metric used as it is the most simplistic, commonly used, and makes the most sense for regression models (JJ, 2016).

Figure 4

Mean Absolute Error Equation

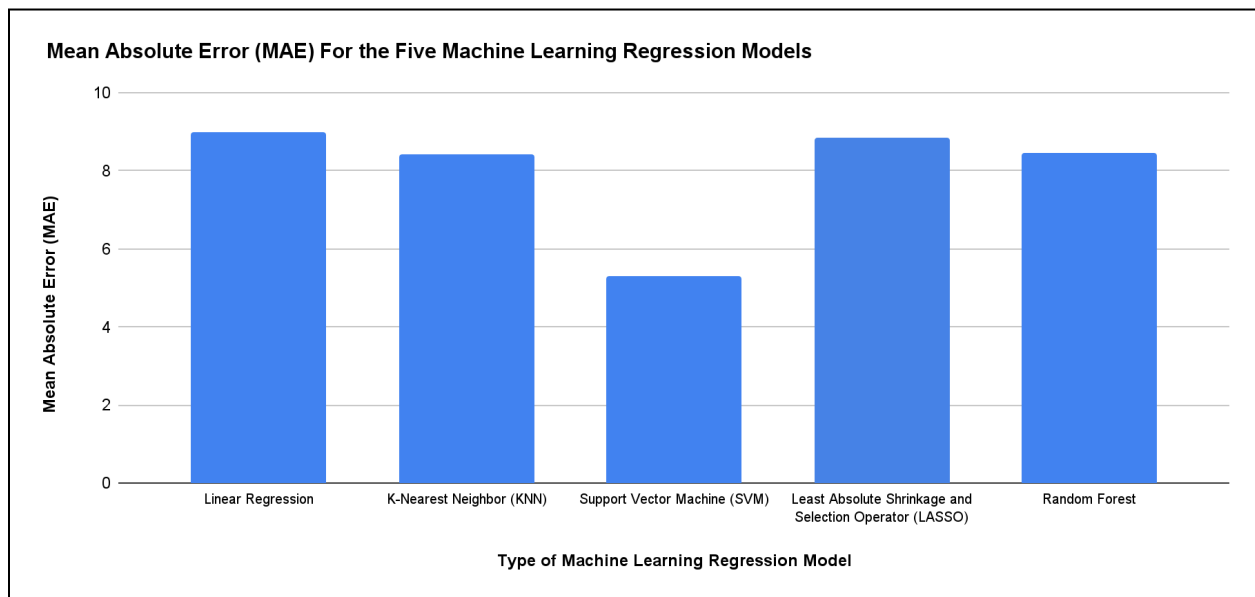
$$\mathbf{MAE} = \frac{1}{n} \sum_{i=1}^n [y_i - y]$$

Note. This equation was used to evaluate the accuracy of the machine learning regression models.

Figure 4 describes the size of the discrepancy between the projected outcome of a prediction and its actual value. MAE calculates the magnitude of errors for a group of predictions and observations by taking the average of absolute errors (C3.ai).

Figure 5

Mean Absolute Error (MAE) for the Five Machine Learning Regression Models



Note. The bar graph demonstrates the accuracy of the five different machine learning regression models by using MAE

Figure 5 shows the MAE scores for all five of the machine learning regression models. As seen in the graph, the MAE for the SVM model is significantly lower than all of the other models, making it the most accurate.

Table 1

MAE, MSE, and RMSE of All Five Regression Machine Learning Models Used

	Linear Regression	KNN	SVM	LASSO	Random Forest
MAE	8.988	8.443	5.289	8.836	8.457
MSE	258.070	229.313	166.394	253.074	237.912
RMSE	16.065	15.143	12.899	15.908	15.424

Note. The table contains values that describe the accuracy of the machine learning regression models.

Table 1 shows MAE, MSE, and RMSE for all five of the machine learning regression models. Though MAE is the main metric being used to evaluate the accuracy of the models, MSE and RMSE are still valuable metrics that assess error. As shown in the table, the SVM model has the least amount of error in all three metrics, making it the model most useful to this study.

Analysis

Model Predictions

Using the SVM model, predictions were able to be made about what particular combinations of gender, race, and mental health status led to the greatest and least number of victims in a mass shooting. By running a code in Python that gives the amount of victims for

every possible combination of inputs, the model declared that the person who would harm the most people is a white woman with noted mental health issues. This was a huge surprise, since 98% of all mass shootings are committed by men (Martin and Bowman, 2021). Given this statistic, it can be assumed that men would also commit the most violent mass shootings. A white woman with mental health issues is predicted to harm just over 8 people in a mass shooting, which was a huge jump from the second highest (7.5 people). On the other hand, the combination of factors that leads to the least amount of victims in a mass shootings is a woman with unknown race and unknown mental health issues. This result was consistent with prior research noted by Martin and Bowman. The findings for the greatest number of victims was surprising as women rarely commit mass shootings. Men are generally regarded as more violent than women and therefore commit more shootings due to a complex combination of evolutionary and psychological variables. Men are more aggressive and less restrained by emotion, and men appear to be less ready to seek aid from others when they are troubled (Park and Howard, 2019).

Permutation Importance

As well as running different machine learning regression models, a permutation importance test was run. Permutation importance describes which factor within a model is the most important and influences the outcome of the test the most. It determines which variables have the greatest influence on the accuracy of a prediction model by rearranging their values and seeing how the performance of the model changes. This test was used to measure which factors have the greatest influence on the number of victims in a mass shooting.

Figure 6

Permutation Importance Test for Support Vector Machine Model

Weight	Feature
0.0321 ± 0.0173	Mental Health Issues_Yes
0.0244 ± 0.0147	Race_Other
0.0211 ± 0.0192	Mental Health Issues_Unknown
0.0112 ± 0.0117	Race_Unknown
0.0092 ± 0.0166	Race_White
0.0048 ± 0.0157	Race_Black
0.0038 ± 0.0044	Gender_Male
-0.0013 ± 0.0053	Race_Latino
-0.0021 ± 0.0034	Race_Native American
-0.0021 ± 0.0017	Gender_Unknown

Note. The permutation importance test demonstrates which aspects of a person have the greatest influence on the number of victims.

Figure 6 exhibits which elements of a person have the greatest influence on the magnitude of victim numbers. Based on the figure, a person having mental health issues has the greatest impact on the amount of people they would harm. If a person has mental health issues, they are more likely to harm 0.0494 people than someone without noted mental health issues. Though this value is not large, it still has an effect in a large dataset. This finding goes against recent studies in the area. Accounts in the media of mass shootings by mentally ill people pique public interest and perpetuate the prevalent notion that mental illness frequently leads to violence. However, According to research, the vast majority of people with severe mental problems are never aggressive (Swanson et al. 2015). Again, the permutation importance test

was implemented to measure which variable had the greatest influence on the amount of predicted victims a mass shooting would have.

Discussion & Conclusion

This study dove into how different factors affect the number of victims in a mass shooting. These factors were the race, gender, and mental health status of the perpetrator. Using five different machine learning regression models (Linear Regression, KNN, SVM, LASSO, and Random Forest), this was able to be tested. SVM turned out to be the most accurate model, with an MAE score of around 5. This model was then used to find which “type” of person would harm the greatest number of people if they were to perpetuate a mass shooting. By running an algorithm that tries every combination of factors, it was determined that the person who would commit the most deadly mass shooting is a white woman with noted mental health issues. On the other side, the person who is predicted to hurt the least amount of people is a woman with unknown race and unknown mental health issues. In addition, a permutation importance test was run in order to find which variable has the greatest impact on the number of victims. It was found that having a mental illness increases the victim count by more than any other factor. This research is ethically beneficial as it helps researchers understand what demographic of people is most likely to harm a large number of people. The model also does not require massive amounts of information to train it, making it extremely efficient in its calculations. By taking into account only three variables, the model is much faster and more applicable in different scenarios. Understanding what factors go into a deadly mass shooting has the potential to save lives in the future.

Limitations of the Study

From combing through many datasets, it can be concluded that mass shootings have only been properly recorded over the past 5-8 years. Before 2015, there are few accounts of mass shootings, and they do not have nearly as much detail as the more recent ones. On top of that, there were even less datasets that focused on the mental health status, race, and gender of the assailants. Because of this, the dataset that was used in the study only contained around 400 mass shooting accounts. Though this is not terrible, it does mean that the model is not as correct as it could be if it had access to more information. The model is only able to base its predictions off of 400 previous data points, when there have been thousands of mass shootings in reality. In addition, there are countless factors that cause a mass shooting to be deadly. Race, gender, and mental health status may be good indicators of how many victims there will be, there are other variables in the equation. Some of these factors include time of day, location of the shooting, number of people at that particular location, type of firearm, and many more. With the datasets being rather simplistic, not taking into account many of these variables, that equates to the machine learning model not being able to take them into account as well. Again, many of the limitations of this study stem from the simplicity of the datasets and lack of recorded information overall.

Future Pursuits

This research has opened my eyes to many different ethical dilemmas in the United States. Learning about the heartbreaking number of mass shooting victims there are every year made me realize how much there is to fix. In addition to the factors explored in this study, elements such as poverty, systemic inequalities, and inequitable access to mental health services also influence the number of victims in a mass shootings. If there was more data on the topic, I

would be interested in further exploring the extent to which the machine learning models can be used to draw conclusions about the lethality of mass shootings.

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